Fourth-Moment Control of the Wilson Nonlinear Correction

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Abstract

For the mass-scaled Wilson–Dirac interval covariance, we derive a vanishing bound for the transverse nonlinear correction, sharpening a square-root-logarithmic upper bound. The argument combines the proven positive spectral-majorant tail estimate with the sine-kernel/Jacobi second moment. Specifically, the transverse contribution is $O_\mu(\sqrt{\log N/N})$, while the longitudinal contribution remains $O_\mu(1)$. Consequently the complete nonlinear correction is uniformly bounded at fixed scaled mass. Together with a separately established cross-boundary entropy bound, this reduces the leading relative-entropy problem to its exact reference-modular residual, without proving the latter vanishes or has a prescribed asymptotic order.

1. Model and inputs

Let $N = 2L$ and $m = \mu/L$ with fixed $\mu > 0$. Consider the interval compression of the occupied-band projection of

$$h_m(k) = \sin k\, \sigma_x + (1 + m - \cos k)\sigma_z.$$

Under the established massless scalarization, the reference covariance is B_N, with commuting Jacobi matrix J_N. Write $W_N = B_N(I - B_N)$, and decompose the perturbation as $\Delta_N = S_N + D_N$. Let v_j be common eigenvectors of B_N and J_N, with eigenvalues $b_j \in (0, 1)$ and λ_j; put $q_j = \langle v_j, G_N v_j\rangle \geq 0$.

The already proved Wilson spectral-majorant theorem supplies

$$|\langle v_j, S_N v_j\rangle| \leq q_j, \qquad |\langle v_j, D_N v_j\rangle| \leq 2\sqrt{q_j b_j(1 - b_j)},$$
$$\sum_{|\lambda_j| > t} q_j \leq \frac{a_\mu}{t} + \frac{b_\mu}{t^2}, \qquad a_\mu = \sqrt{\frac{2}{15}}\frac{\mu^2}{\pi}, \quad b_\mu = \frac{\mu}{3\pi}, \qquad t > 0.$$

The Jacobi normalization ensures $|\lambda_j| \leq N/2$. The exact half-filled sine-kernel boundary identity implies

$$0 \leq M_{2,N} := \mathrm{Tr}(J_N^2 W_N) \leq \frac{32}{\pi^2}(H_N + 1),$$

and the refined exterior-lattice computation proves $M_{2,N} = (3\pi^2)^{-1} \log N + O(1)$.

Define $\eta(t) = \pi t\, {}_3F_2(\frac{1}{4}, \frac{1}{2}, \frac{3}{4}; 1, \frac{3}{2}; 4t^2)$ and $V_N = -N[\eta(J_N/N) - \pi J_N/N]$. The positive-coefficient hypergeometric series gives

$$|\eta(t) - \pi t| \leq C_\eta |t|^3 \quad (|t| \leq \tfrac{1}{2}), \qquad C_\eta = 4\pi\big[{}_3F_2(\tfrac{1}{4}, \tfrac{1}{2}, \tfrac{3}{4}; 1, \tfrac{3}{2}; 1) - 1\big].$$

All inputs are specific to the Wilson–Dirac/sine-kernel model and do not assume a reference-modular comparison.

2. A fourth-moment bound for the positive majorant

Lemma 1. For every even $N \geq 2$,

$$\boxed{\mathrm{Tr}(G_N J_N^4) = \sum_j q_j |\lambda_j|^4 \leq \frac{a_\mu N^3}{6} + \frac{b_\mu N^2}{2}.}$$

Proof. The diagonal representation holds because J_N^4 is diagonal in the common reference eigenbasis, regardless of whether G_N commutes with J_N. For the positive measure $\nu_N = \sum_j q_j \delta_{|\lambda_j|}$, layer-cake integration and the Jacobi spectral radius give

$$\begin{aligned}\mathrm{Tr}(G_N J_N^4) &= 4 \int_0^{N/2} t^3 \nu_N((t,\infty))\, dt \\ &\leq 4 \int_0^{N/2} (a_\mu t^2 + b_\mu t)\, dt = \frac{a_\mu N^3}{6} + \frac{b_\mu N^2}{2}.\end{aligned}$$

The singular appearance of the spectral-tail bound near $t = 0$ causes no problem: the integrated majorant is finite. □

3. Improved transverse estimate

Theorem 2. Let $\mathcal{Q}_N^{\mathrm{trans}} = \mathrm{Tr}(D_N V_N)$. Then

$$\boxed{|\mathcal{Q}_N^{\mathrm{trans}}| \leq 2C_\eta \sqrt{\left(\frac{a_\mu}{6N} + \frac{b_\mu}{2N^2}\right) M_{2,N}} = O_\mu\left(\sqrt{\frac{\log N}{N}}\right).}$$

In particular $\mathcal{Q}_N^{\mathrm{trans}} \to 0$ as $N \to \infty$ at each fixed $\mu > 0$.

Proof. In the reference eigenbasis, V_N has diagonal entries w_j with $|w_j| \leq C_\eta |\lambda_j|^3 / N^2$. Using the pairing inequality followed by Cauchy–Schwarz with the **fourth positive-majorant moment**, not merely its total mass, yields

$$\begin{aligned}|\mathcal{Q}_N^{\mathrm{trans}}| &\leq \frac{2C_\eta}{N^2} \sum_j |\lambda_j|^3 \sqrt{q_j b_j (1 - b_j)} \\ &\leq \frac{2C_\eta}{N^2} \left(\sum_j q_j \lambda_j^4\right)^{1/2} \left(\sum_j \lambda_j^2 b_j (1 - b_j)\right)^{1/2} \\ &= \frac{2C_\eta}{N^2} \sqrt{\mathrm{Tr}(G_N J_N^4)\, \mathrm{Tr}(J_N^2 W_N)}.\end{aligned}$$

Apply Lemma 1 and the established Jacobi moment bound. □

The finite-length estimate is explicit; the sharp asymptotic coefficient for $M_{2,N}$ is not required to obtain the rate, only its $O(\log N)$ upper bound.

4. Bounded nonlinear correction and modular reduction

The established longitudinal estimate is

$$|\mathcal{Q}_N^{\mathrm{long}}| \leq C_\eta \left(\frac{3a_\mu}{8} + \frac{3b_\mu}{2N}\right).$$

Therefore

$$\boxed{|\mathcal{Q}_N| \le C_\eta \left(\frac{3a_\mu}{8} + \frac{3b_\mu}{2N}\right) + 2C_\eta \sqrt{\left(\frac{a_\mu}{6N} + \frac{b_\mu}{2N^2}\right) M_{2,N}} = O_\mu(1).}$$

The exact two-bond asymptotic and finite quasifree relative-entropy decomposition are

$$\mathcal{P}_L = \frac{\mu^2}{6} \log L + O_\mu(1), \qquad \mathcal{D}_L = \mathcal{P}_L + \mathcal{Q}_N + \mathcal{F}_L - \Delta S_L,$$

where $\mathcal{F}_L = \mathrm{Tr}[(A_N - B_N)E_N]$ is the exact reference-modular residual. Theorem 1 of *Uniform Square-Root-Mass Control of Wilson–Dirac Boundary Covariances*, with its entropy argument in section 5, proves $\Delta S_L = O_\mu(1)$ for this Wilson scaling. Combining that imported bound with the nonlinear result of this paper yields

$$\boxed{\mathcal{D}_L = \frac{\mu^2}{6} \log L + \mathcal{F}_L + O_\mu(1).}$$

This reduction alone does not prove $\mathcal{F}_L = o(\log L)$, convergence to a finite constant, or a proposed inverse-logarithmic convergence rate. The separate theorem in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* proves $\mathcal{F}_L = O_\mu(1)$, sufficient for the leading law. Theorem 3 of *Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals* proves $\|E_N\| \le 7$ for every even $N \ge 2$, under the exact concentration connection specified there. This later result is separate from the reduction proved above.

5. Evidence and scope

The fourth-moment estimate and theorem are exact mathematical consequences of the stated majorant and sine-kernel results. Independent finite arithmetic compares the improved bound with the earlier bound for eight Wilson configurations; all satisfy the stronger estimate. These tests are consistency checks, not the proof. The claimed rate holds at fixed μ only, with no uniform assertion for μ varying with N.

External mathematical context

- V. Eisler and I. Peschel, "Analytical results for the entanglement Hamiltonian of a free-fermion chain," *J. Phys. A* **50** (2017) 284003, doi:10.1088/1751-8121/aa76b5.
- D. Slepian, "Prolate spheroidal wave functions, Fourier analysis, and uncertainty. V: The discrete case," *Bell System Technical Journal* **57** (1978), 1371–1430, doi:10.1002/j.1538-7305.1978.tb02104.x.
- S. Karnik, J. Romberg and M. A. Davenport, "Improved bounds for the eigenvalues of prolate spheroidal wave functions and discrete prolate spheroidal sequences," *Applied and Computational Harmonic Analysis* **55** (2021), 97–128, doi:10.1016/j.acha.2021.04.002.

These references support the classical concentration-operator setting; they do not independently prove the new Wilson-specific fourth-moment consequence.

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Kirtus Justus directed the research. ChatGPT (OpenAI) developed and audited this fourth-moment derivation and its computational corroboration. Claude (Anthropic), Gemini (Google), Grok (SpaceXAI), and DeepSeek (High-Flyer) are acknowledged as broader research contributors without attribution of this derivation to them. There has been no external human peer review or institutional endorsement.